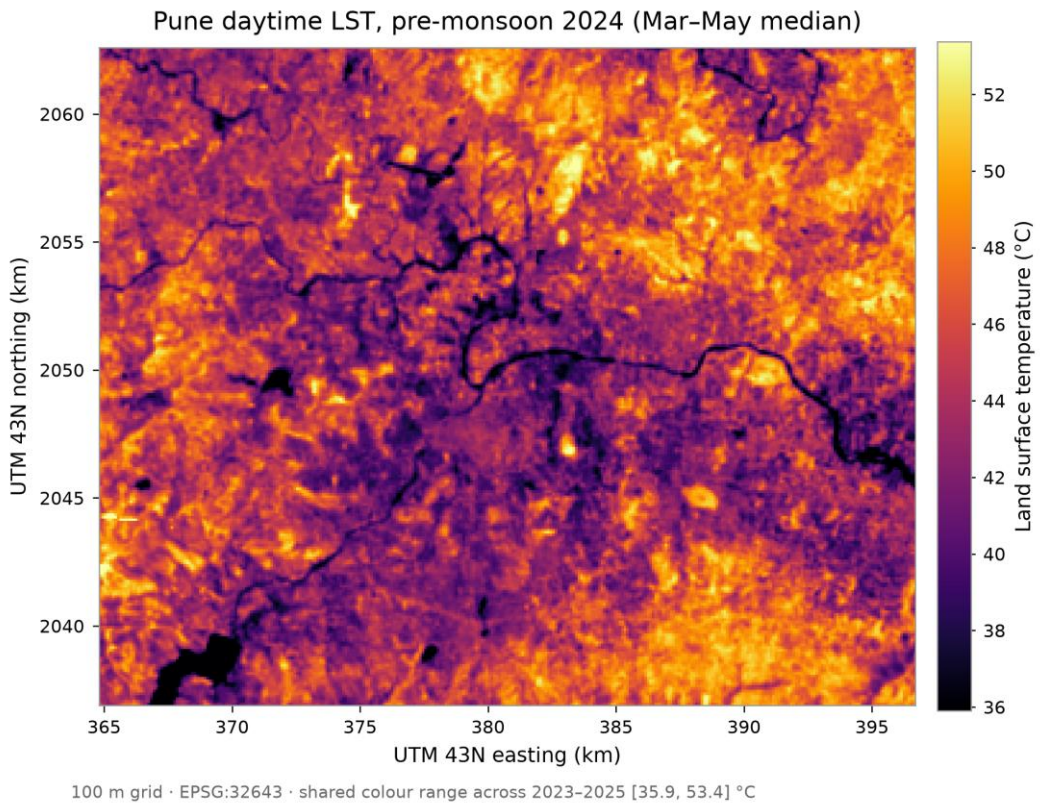
MirageAI

Concrete vs. Canopy: Explainable Machine Learning for Urban Heat Island Drivers in Pune

Shourya Tiwari · Shashwat Kumar
Rishik Sinha · Shubhankar Bhan



Applications and Use Cases of Machine Learning (TE7483)
Symbiosis Institute of Technology, Pune

Where would greening cool Pune most?

3.4–4.0 °C

cooler: city core vs outer ring, land cells, daytime, 2023–25

- Heat plans ask where greening cools most.
- Satellite LST: free, city-wide, repeatable.
- Assume a hot core, target the wrong places.
- Random-pixel validation flatters models.

PROBLEM STATEMENT

Predict and explain surface heat where the model has never looked

Predict each 100 m cell's daytime LST anomaly in unseen places and years, and explain it.

Honest accuracy

Unseen areas and an unseen year.

Trustworthy explanation

Stable across models and methods.

Bounded scenarios

Ranges, not forecasts.

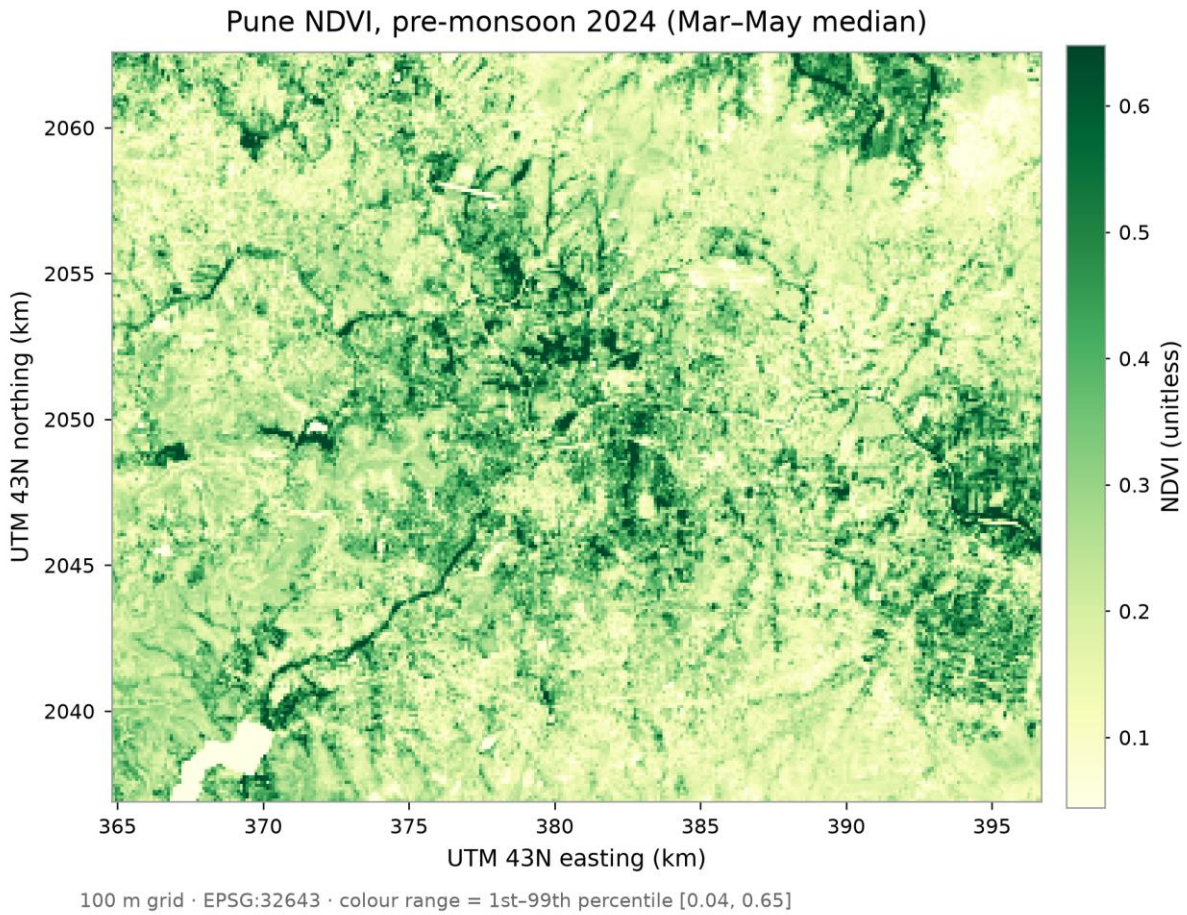
Hard rule: coordinates are never model inputs.

Objectives: five research questions

RQ1	**Drivers:** Which land-cover and terrain factors go with daytime LST?
RQ2	**Nonlinearity:** Do tree models find thresholds and interactions that linear models miss?
RQ3	**Validation:** How much does random CV overstate accuracy?
RQ4	**Greening:** How much cooling goes with more tree cover in the hottest cells?
RQ5	**Cool island:** Is the built core cooler by day, and is that the buildings?

DATASET

Four open datasets on one 100 m grid



100 m grid · EPSG:32643 · colour range = 1st–99th percentile [0.04, 0.65]

Landsat 8/9 surface temperature (target)

Sentinel-2 NDVI, dryness, moisture

WorldCover 2021 7 land-cover fractions

SRTM elevation, slope

100 m grid · 3 seasons · 243,252 land cell-years

METHODOLOGY

Validation designed before any model is tuned

Target

anomaly from each year's median

Block size

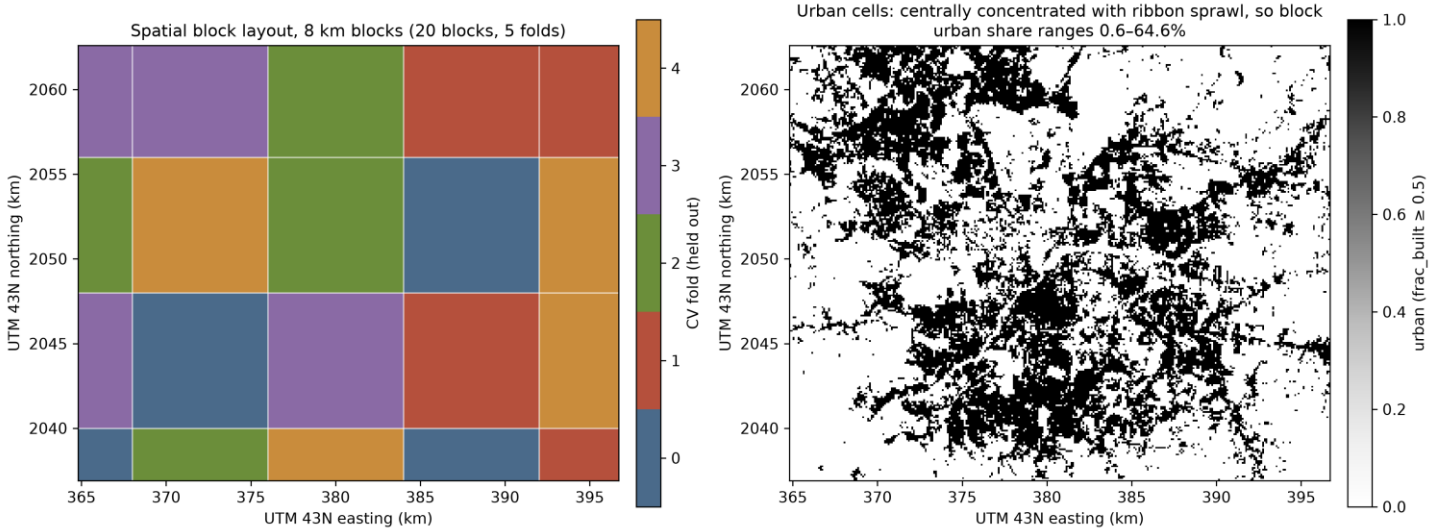
residual range 3,950 m → 4 km

Layouts

5 random seeds × 5 folds

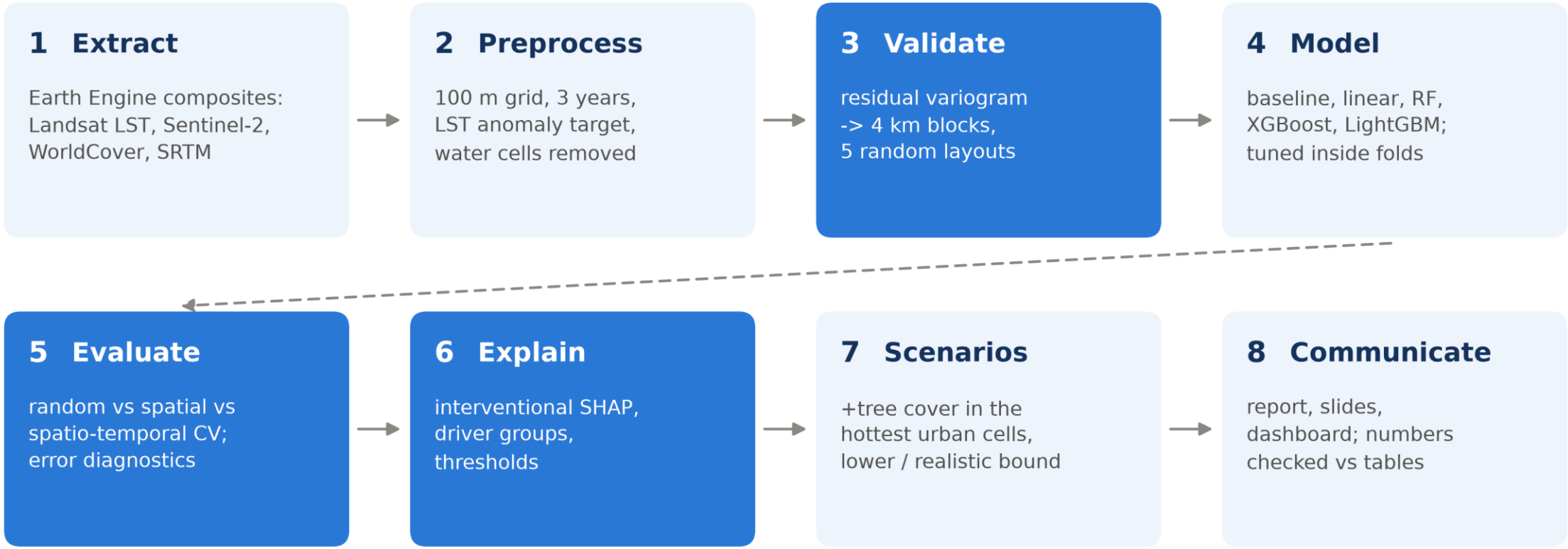
Final test

2025, unseen blocks, untouched



ML PIPELINE

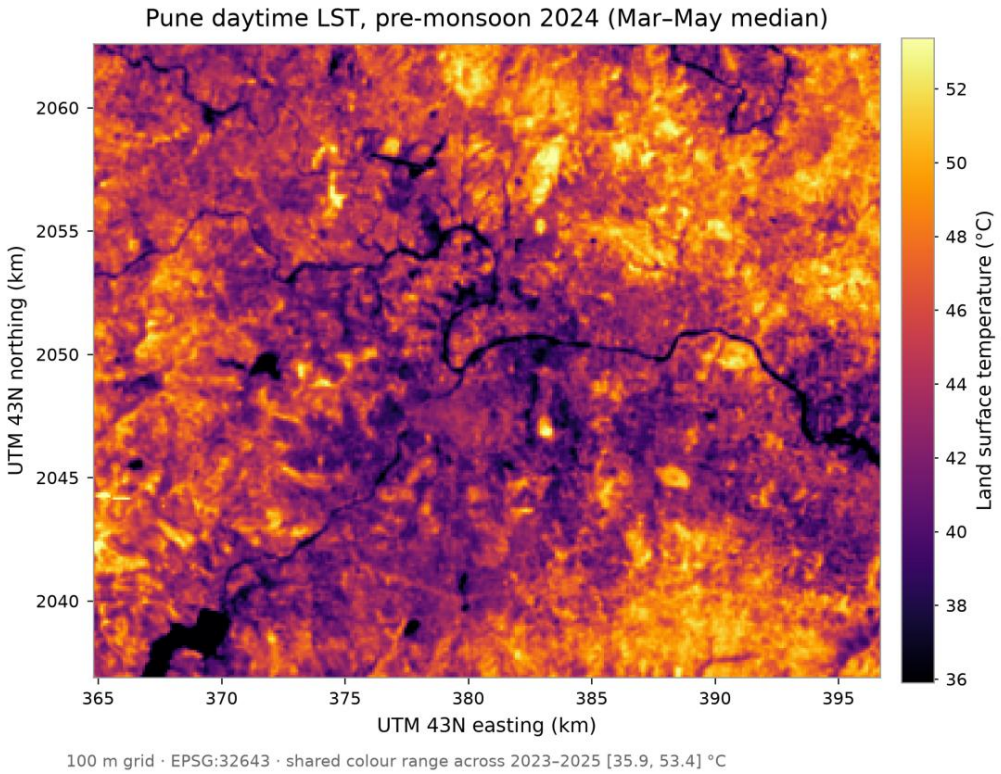
Eight stages, one command



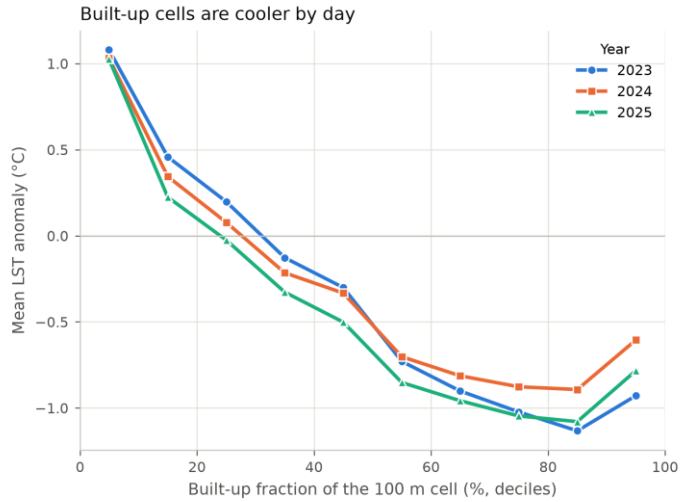
blue: the steps that decide whether the results can be trusted

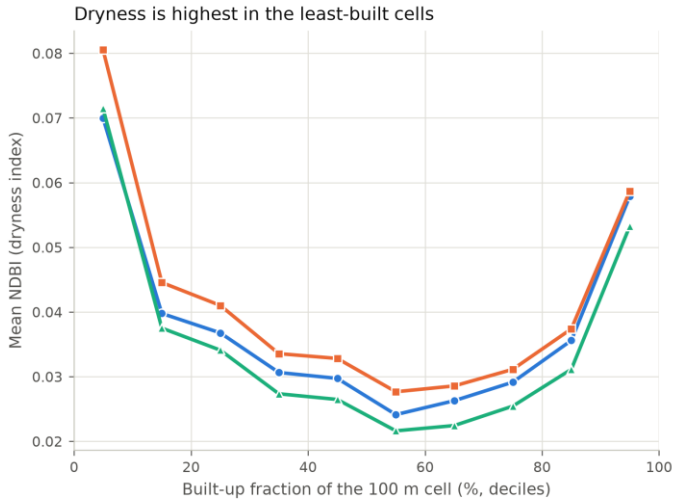
One command rebuilds every table and figure; a script re-checks every quoted number.

The mirage: the built core is cooler by day



100 m grid · EPSG:32643 · shared colour range across 2023–2025 [35.9, 53.4] °C





Land cells, pre-monsoon (Mar-May) composites, 2023-25. Anomaly = cell LST minus that year's AOI median. One line per year.

2.1 °C

least-built vs most-built cells

NDBI peaks in the least-built cells: here it measures dryness.

MODELS

A baseline and four learners on identical folds

Baseline

training mean: the floor

Linear

standardised in-fold; the status quo

Random Forest

bagged deep trees

XGBoost

boosted trees, depth-wise

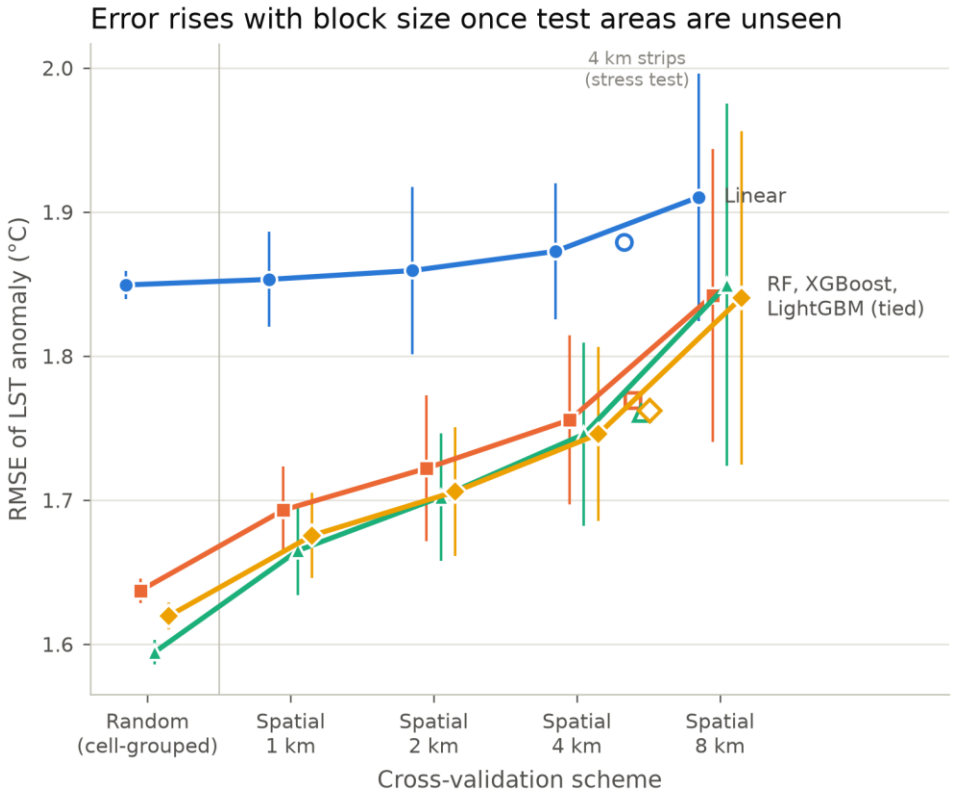
LightGBM

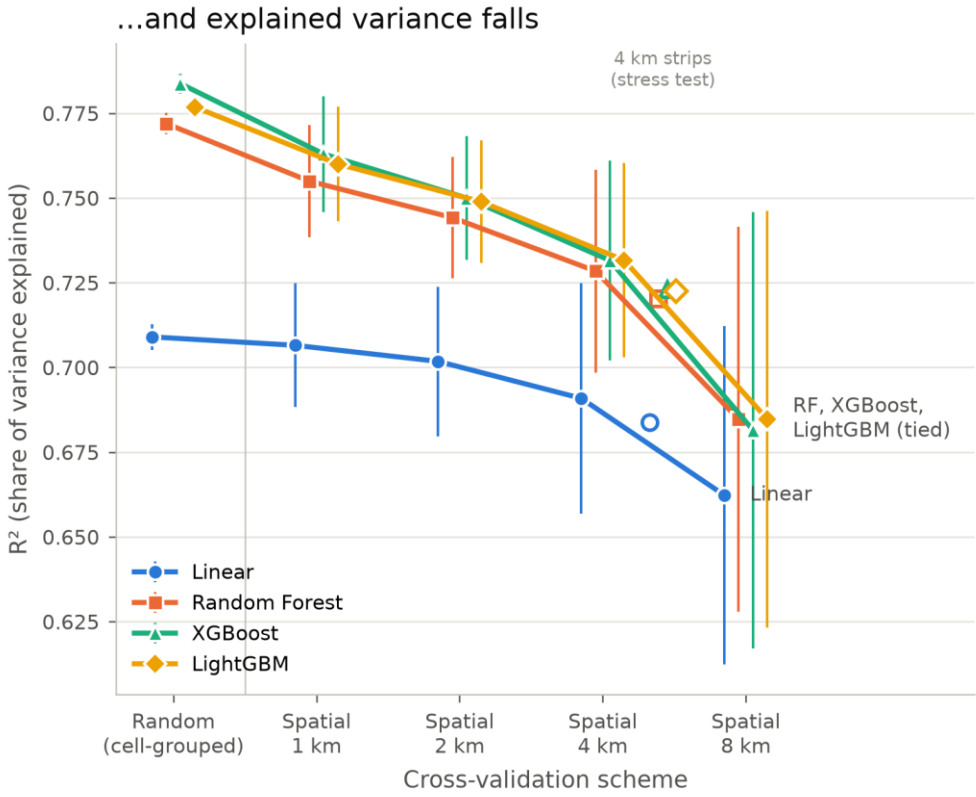
boosted trees, leaf-wise

Tuning, inside spatial CV

Randomised search · 25 candidates per model · 4 km spatial folds · 2023 and 2024 only · 60,000-row sample

Random cross-validation flatters flexible models





Mean ± std over 5 seeds × 5 folds; water-majority cells excluded (D-034). Hollow markers: unshuffled 4 km layout, every fold an east-west strip (D-033).
Baseline (RMSE 3.39-3.49 °C, $R^2 \approx 0$) omitted for scale. Linear uses 12 features, trees 13 (D-025).

Random → 4 km blocks: linear +0.02 °C, trees +0.12–0.15 °C. Flexible models are flattered most.

The three tree models tie once places are unseen

RMSE, °C	Random CV	4 km blocks	2025 unseen	2025 seen place
Linear	1.85	1.87	1.90	1.88
Random Forest	1.64	1.76	1.75	1.52
XGBoost	1.59	1.75	1.75	1.56
LightGBM	1.62	1.75	1.75	1.60

1.75–1.76 °C

trees tied, 4 km blocks

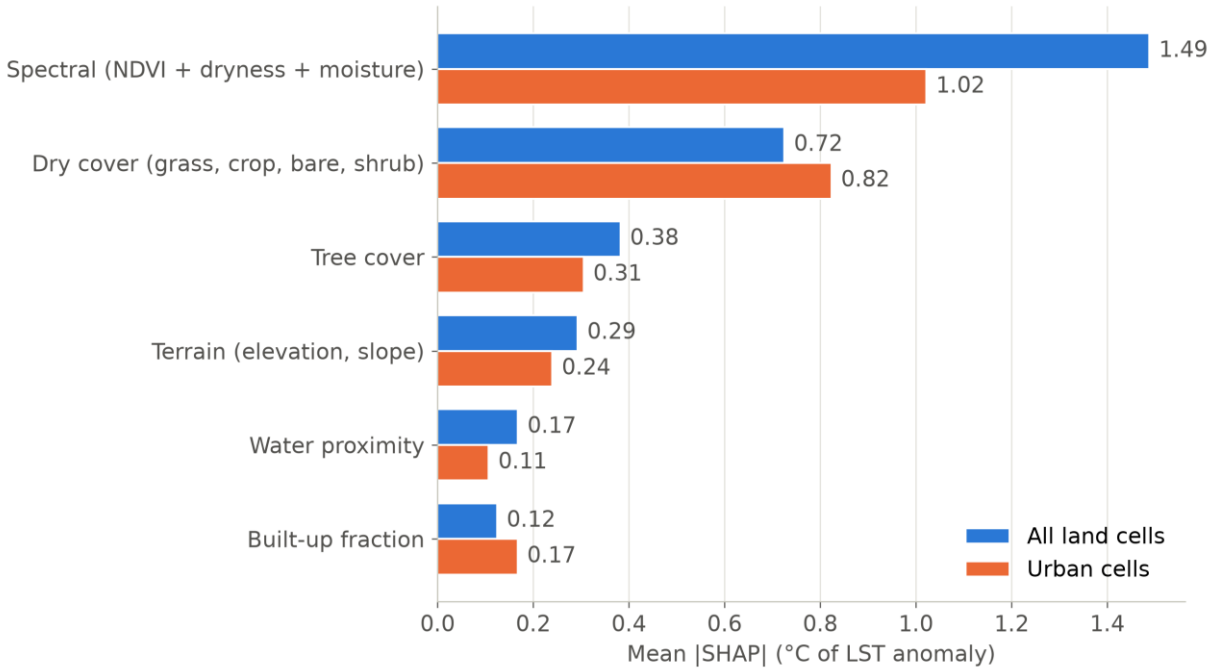
R^2 0.72

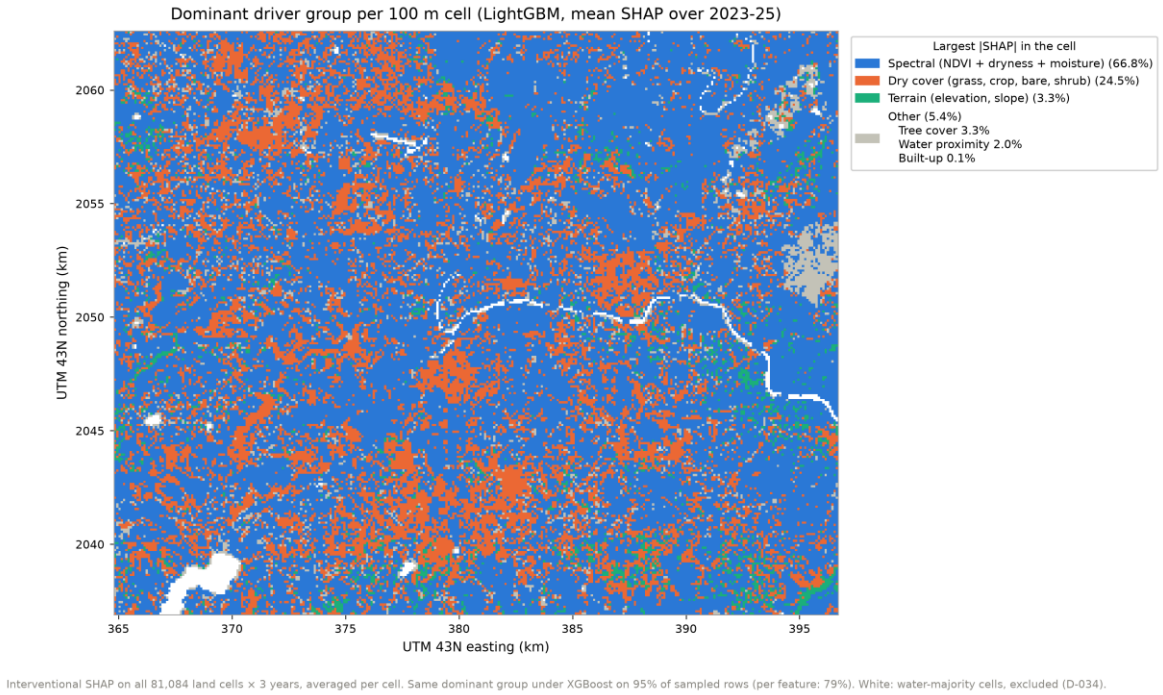
2025, unseen places

0.15–0.23 °C

memorisation gap, trees (linear 0.02)

Vegetation, dryness and moisture lead





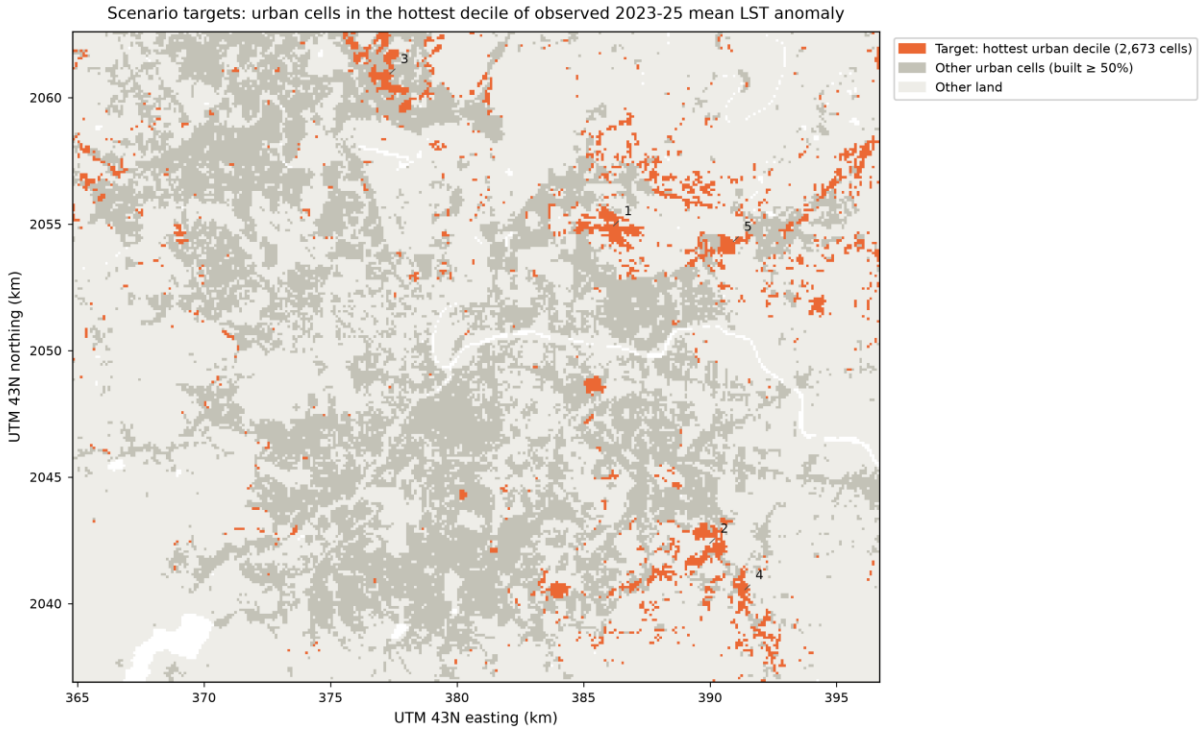
Interventional SHAP on all 81,084 land cells × 3 years, averaged per cell. Same dominant group under XGBoost on 95% of sampled rows (per feature: 79%). White: water-majority cells, excluded (D-034).

1.49 °C
spectral block, mean |SHAP|

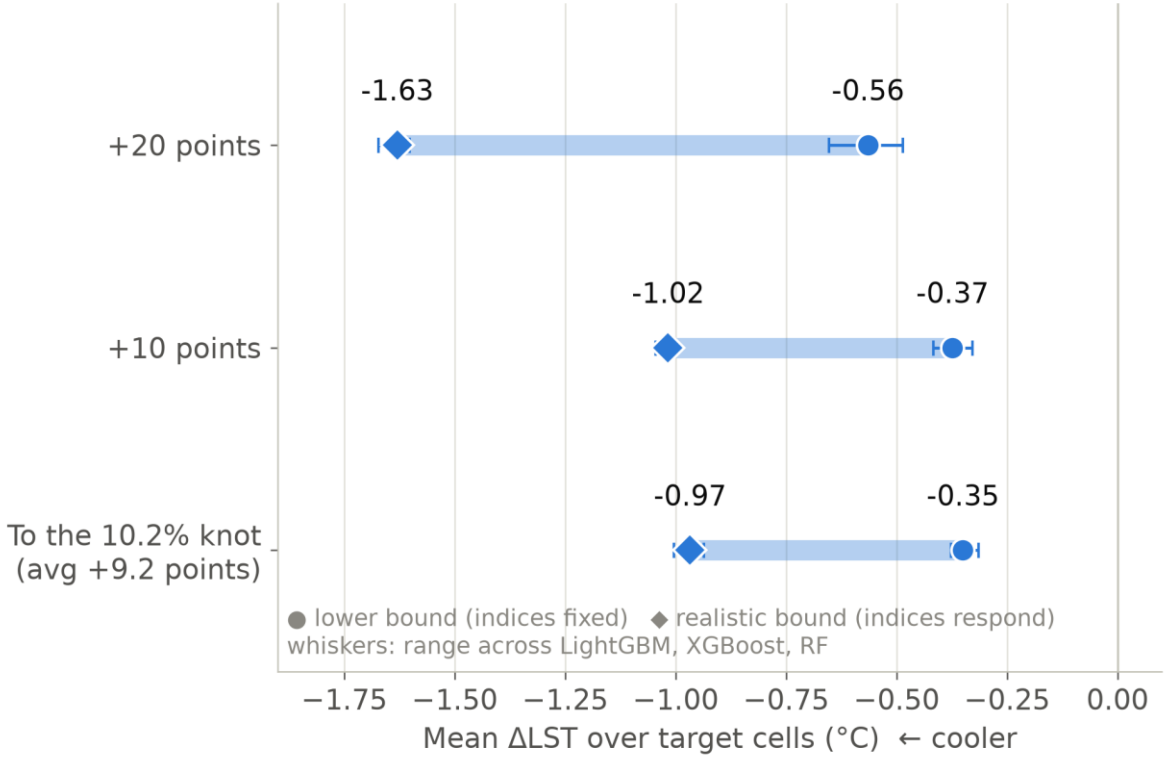
2.1×
the dry-cover group

95%
same leading group, XGBoost

Greening the hottest urban cells



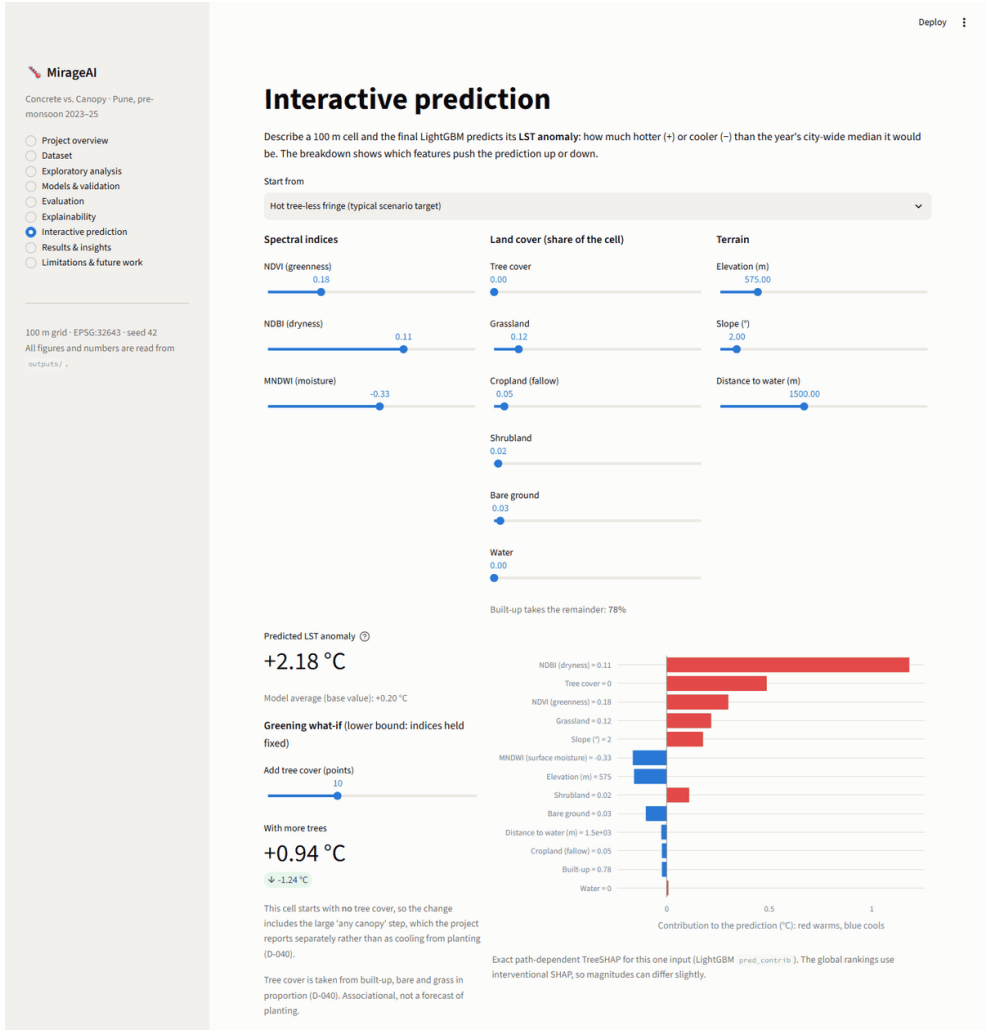
Numbers: the five largest clusters of 8-connected target cells. Threshold +2.41 °C. White: water-majority cells, excluded (D-034).



~10% canopy: 0.4–1.0 °C cooler. Gains per point halve past ~10–12%. First-pixel step excluded.

DASHBOARD DEMO

Every result, explorable in a browser



Run it

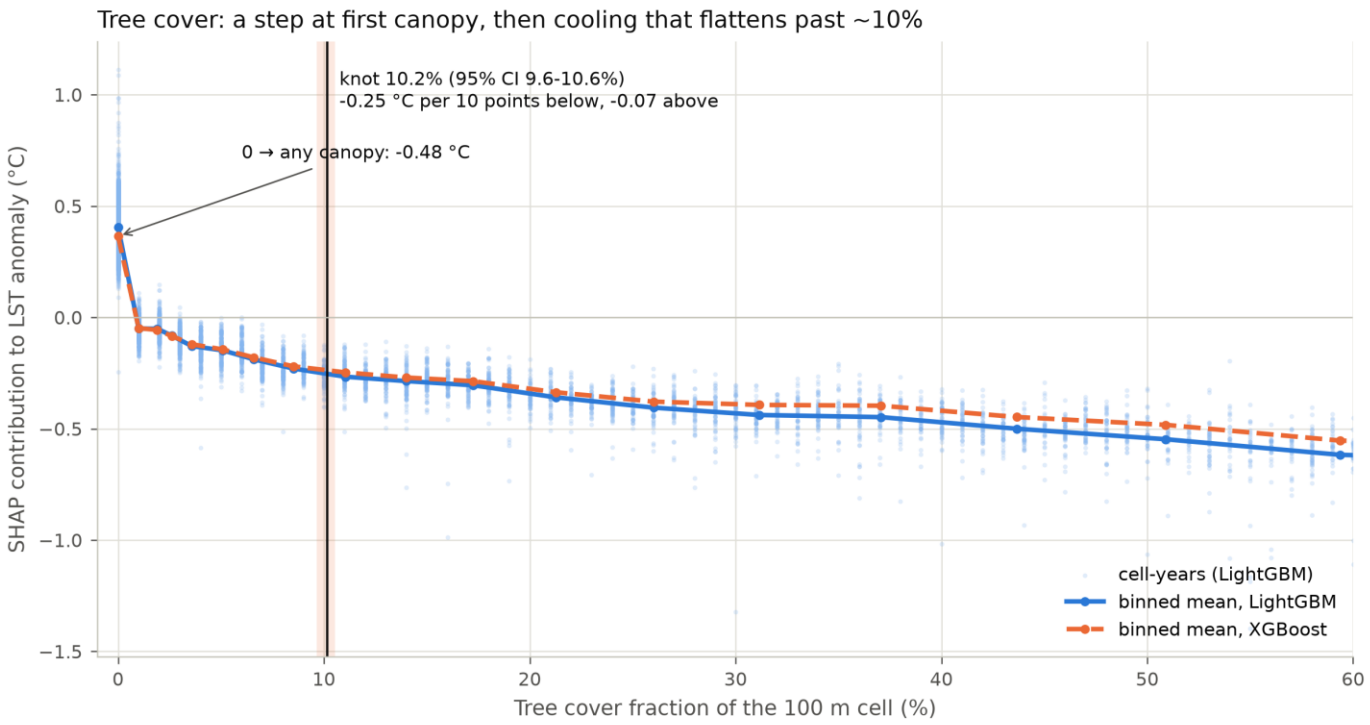
```
streamlit run app.py
```

Demo path

- Pick a hot, tree-less fringe cell.
- Read its per-feature breakdown.
- Add tree cover: lower-bound what-if.
- Read-only: numbers come from the pipeline.

KEY INSIGHTS

Concrete vs canopy: what the evidence says



Interventional SHAP, 10,000 stratified cell-years. Knot and slopes: segmented fit on cells with tree cover > 0 (D-037). Associational, not causal.

−87%

built-up association once vegetation and dryness are modelled

10.2%

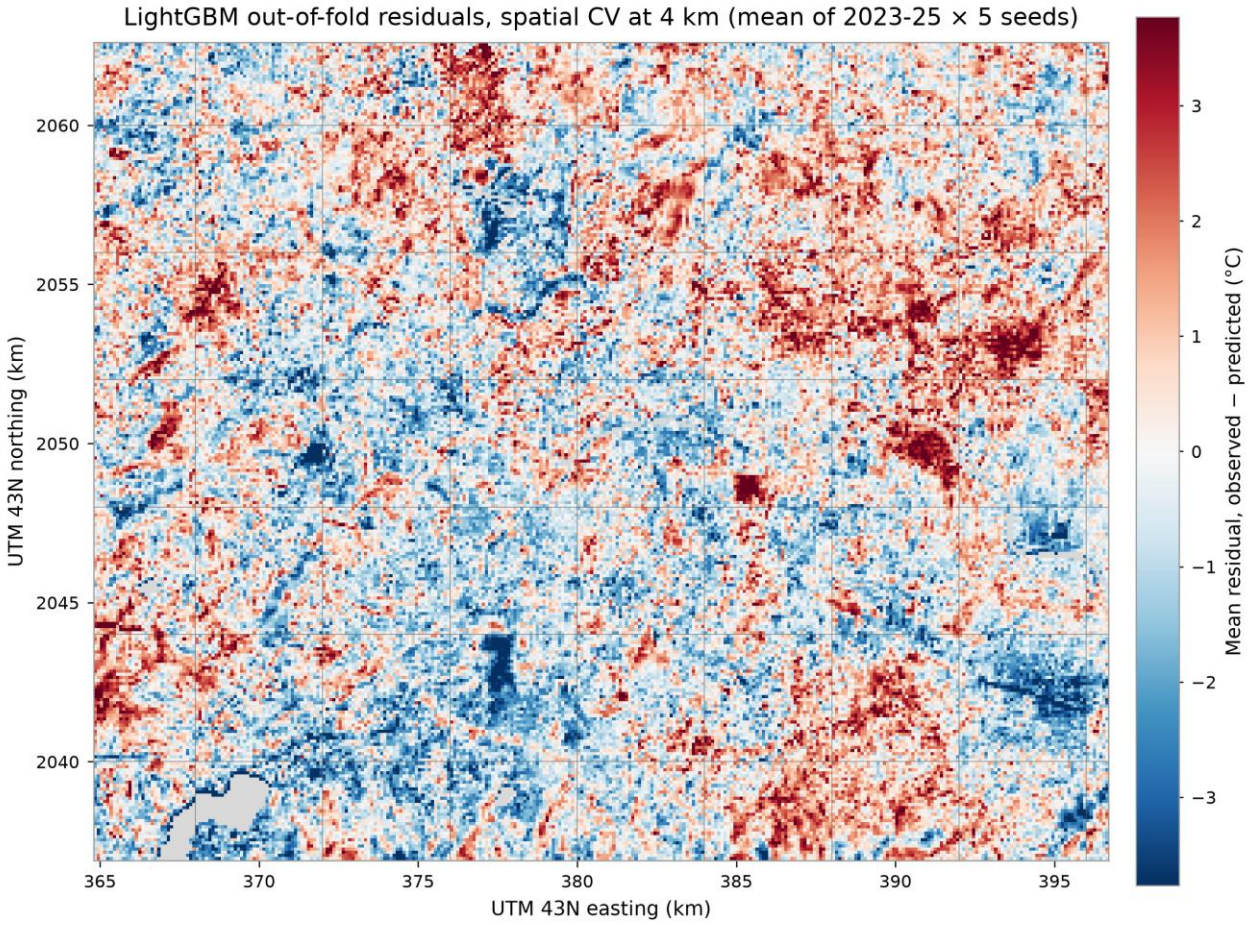
tree-cover knot: gains per point flatten beyond

Spatial CV

the honest score; random CV flatters flexible models

What these results cannot tell us

- Daytime surface LST, not air temperature
- Associations, not causal effects
- Residual Moran's I ≈ 0.6: missing drivers
- Static 2021 land cover
- One city, one season



FUTURE SCOPE

Where the work goes next

Urban form

building height, roofs, irrigation

Air temperature

ground and mobile sensors

Before and after

multi-year land-cover change

Night and seasons

where the island may flip

Nested spatial CV

remove tuning optimism

Live scenarios

realistic bound in the dashboard

Conclusion

1.75 °C
RMSE, unseen places and year

R^2 0.72
on the 2025 test

0.4–1.0 °C
cooler at ~10% canopy

Pune's hottest urban surfaces are tree-sparse fringe developments, mostly on the eastern edge. In these data, raising them to ~10% tree cover is associated with ~0.4–1.0 °C cooler daytime surfaces, with gains per point roughly halving beyond ~10–12% cover. These are cross-sectional associations, not measured effects of planting.

Thank you

Questions?

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Report · slides · dashboard · code: MirageAI repository